Nuclear Hormone Receptor Signaling

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Block: Endocrine/Urology/Reproduction

Thread: Biochemistry

Conflict of Interest: Dr. Scott has no conflicts of interest to disclose.

How this Lesson fits into the Curriculum

In the BCM thread so far, you have learned about predominantly hydrophilic molecules signaling via receptors that are integral membrane proteins (lessons in Introduction to the Study of Medicine, Host Defense/Blood, Pulmonology and in last week's lesson on Regulation of Plasma Glucose, as well as various CBL sessions). In this lesson, you will learn about membrane-permeable signaling molecules that bind to intracellular receptors (nuclear hormone receptors) that function primarily through regulation of transcription.

One example of cell-permeable signaling molecules (hormones) is steroids, which regulate the immune system (Host Defense/Blood), blood pressure (Cardio, Renal, and this block), proliferation of endocrine-responsive tissues (e.g., breast cancer (Pulmonary) and prostate cancer (this block), glucose metabolism, sexual development & reproduction (this block), as well as bone mineral metabolism (GI & MSK).

Several steroid drugs are on the WHO List of Essential Medicines. You already know how steroids are used to treat allergies, asthma, and IBD, for instance.

In this block, the content of this lesson applies to the following:

Thyroid hormone signaling in CBL in Week 2.

Testosterone signaling in CBL in Week 3.

Steroid hormone synthesis and signaling in CBL in Week 3.

Estrogen signaling in CBL in Week 4 and a PHARM lesson entitled Estrogens and Progestins, also in Week 4.

Learning Objectives

By exam time, you should be able to do the following:

1. Discuss how steroid and non-steroid hormones pass through the plasma membrane to access nuclear hormone receptors.
2. Compare and contrast type I and type II nuclear hormone receptor signaling.
3. Describe the role of heat shock proteins in steroid signaling.
4. Summarize the typical pathway through which steroid signals induce an increase or decrease in transcription of select genes.
5. Discuss how nuclear hormone receptors that recognize non-steroid ligands regulate transcription.
6. Explain how cell- or tissue-specific effects on nuclear hormone signaling are achieved.
7. Outline the mechanism of action of peroxisome proliferator-activated receptors (PPARs) and discuss the utility of these receptors in the management of dyslipidemia and insulin resistance.

Study Questions

1. Create a table that describes the differences in the mechanism of action of the estrogen receptor and the thyroid hormone receptor.

2. The estrogen receptor is a member of the nuclear receptor superfamily and is important for mediating the effects of estrogen in both XX and XY individuals. The estrogen receptor, like many receptors in this family, resides in the cytoplasm in the absence of the hormone estradiol and translocates to the nucleus upon binding hormone.

   Once in the nucleus, how does the estrogen receptor lead to transcriptional activation of its target genes? Please start with its binding to DNA and finish with successful transcription initiation by RNA Polymerase II.

3. A 9-year-old patient was referred for evaluation of palpable gonads in the inguinal region. At birth, bilateral inguinal gonads with blind-ended vagina were detected. The child was followed over a period of 7 months, after which assignment of female sex was made based on the appearance of external genitalia. Surgical removal of inguinal gonads had been planned, but was not carried out. During the current visit, pelvic sonogram did not reveal any Mullerian structures. Hormonal evaluation revealed FSH (5.4 mIU/mL; normal: 1.5–12.4), LH (21.2 mIU/mL; Normal: 1.7–8.6) and total testosterone (15 ng/mL; normal: 2.86–8.1). Karyotype was 46 XY. Testosterone (T) and dihydrotestosterone (DHT) synthesis defects were excluded by the normal rise of T and DHT after HCG stimulation (basal T: 15 ng/ml; basal DHT: 9.3 ng/ml; T/DHT ratio after HCG stimulation: 7.9; and normal post-HCG stimulation, T/DHT ratio is <17). Surgical removal of inguinal gonads was performed a few months later; histologic analysis revealed normal bilateral testes. The patient subsequently underwent vaginoplasty and received therapy with estrogens. Genomic DNA was extracted and exonic PCR amplification of the androgen receptor (AR) gene was performed. Sequence analysis revealed a C to T transition in exon 6 of the AR gene, converting a CAG codon to a TAG codon at the beginning of the ligand-binding domain.

   What is your explanation for the complete androgen insensitivity syndrome presenting in this patient?

A. Introduction to nuclear hormone signaling

1. Nuclear hormone receptors bind small-molecule hormones that are membrane permeable; hormone binding elicits effects of nuclear hormone receptors on transcription and, thereby, cellular functions.

2. In this lesson we focus on and distinguish between type I and type II NHRs.
   a. Type I NHRs
      i. located **in cytosol in the absence of hormone**
      ii. translocate to the nucleus upon hormone binding
      iii. form **homodimers** on specific DNA sequences.
      iv. Include **mineralocorticoid receptor** (MR; primary ligand: aldosterone), **glucocorticoid receptor** (GR; corticosterone, cortisol), **androgen receptor** (AR; testosterone and dihydrotestosterone), **estrogen receptor** (ER; estradiol).

   b. Type II NHRs
      i. bound to DNA, even in the absence of hormone
      ii. form **heterodimers with retinoid X receptor** (RXR - RXR homodimers also fall into this class) on specific DNA sequences.
      iii. suppress transcription (usually) in the absence of hormone; activate transcription in presence of ligand.
      iv. include **thyroid hormone receptor** (TR; thyroxine (T4), triiodothyronine (T3)), **vitamin D receptor** (VDR; vitamin D3), and **retinoic acid receptor** (RAR, RXR; retinoic acid).

   c. Peroxisome proliferator-activated receptors
      i. share properties of both Type I & Type II NHRs
      ii. largely **cytosolic** in the absence of ligand
      iii. form **heterodimers with retinoid X receptor** on specific DNA sequences.
      iv. Three types (**PPAR-α, β/δ & γ**; see section H)

B. Steroid hormones

1. Non-polar molecules derived from cholesterol that activate a broad array of physiological functions.

2. Function through the interaction with intracellular receptors, which mediate effects of hormone binding by primarily activating or repressing transcription.

3. The hydrophobicity of steroid hormones makes them insoluble in the bloodstream; they are transported by specific carrier proteins.

C. Steroid hormone signaling

1. Steroid hormone signaling occurs upon the high affinity binding of each steroid hormone to the ligand binding domain of its specific receptor (NHR). For type I and II NHRs, the bound receptor regulates transcription of specific hormone-responsive genes. Specificity of transcriptional regulation is determined by DNA binding to specific response elements within promoter/enhancer regions of target genes.

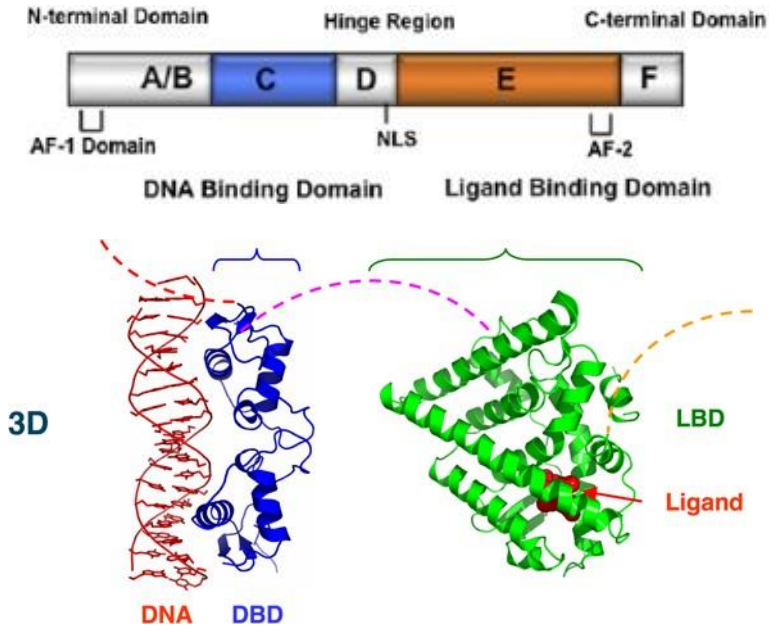
2. Entry of steroidal NHR ligands into the cell

    a. In serum, free and protein-bound steroids are in equilibrium with one another, although protein-bound steroids predominate (> 95%).

    b. Unbound steroid hormones can freely diffuse across the cell membrane; no evidence of hormone uptake while conjugated to a binding protein (e.g., sex hormone binding globulin (SHBG) or corticosteroid binding globulin (CBG)).

    c. Depending on cell and tissue type, and the levels of hormone metabolizing enzymes, steroidal hormones can be converted to other forms within the cell (see section E).

3. NHR domain structure:



AF-1 and AF-2 domains are transcriptional activation domains. NLS refers to the nuclear localization signal.

4. Type I nuclear hormone receptors
   a. Form **homodimers.**
   b. Examples: Glucocorticoid receptor
      Mineralocorticoid receptor
      Androgen receptor
      Progesterone receptor
      Estrogen receptor
   c. Bind primarily to specific DNA sequences within promoters and enhancers of target genes.

d. Bind via zinc finger motif within DNA binding domain.

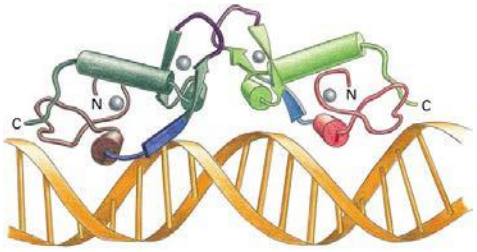


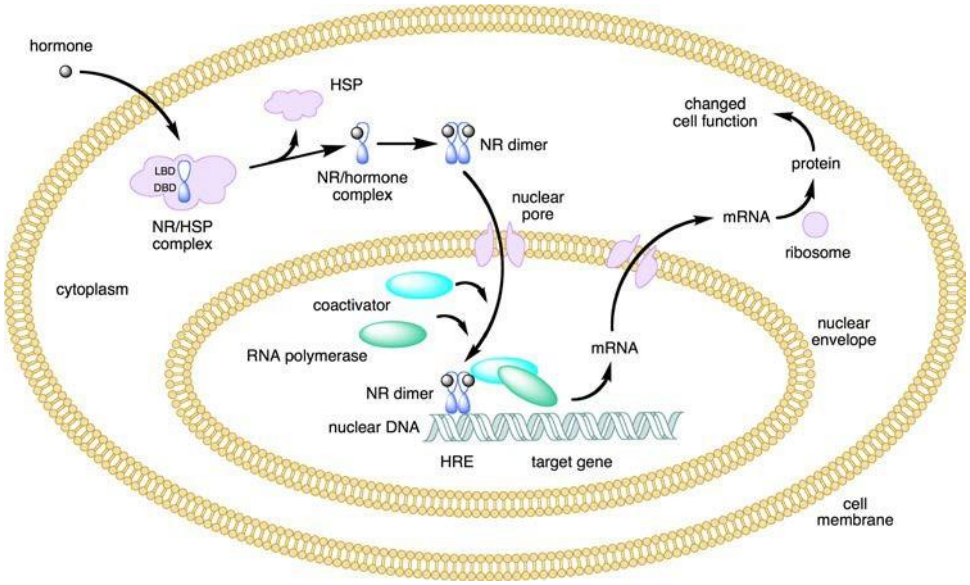
Figure 7–19. Molecular Biology of the Cell, 4th Edition.

e. Cytoplasmic in the absence of hormone

f. Retention of steroid hormone receptors in the cytoplasm requires components of the aporeceptor complex.

   i. Aporeceptor complex – Upon translation, the nuclear receptor interacts with various molecular chaperones (also known as heat shock proteins) and several other accessory proteins, to form the aporeceptor complex, also called the "foldosome."

   ii. Upon hormone binding, the receptor undergoes a conformational change, releasing it from several foldosome components, exposing the nuclear localization signal and stimulating receptor dimerization, nuclear localization, and DNA binding. Some molecular chaperones (e.g., Hsp90) are required for efficient transport of NHRs into the nucleus.



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D. Activation of transcription by NHRs

1. Nuclear receptors regulate transcription by first binding to specific DNA response elements within gene promoters and/or enhancers according to receptor type. Keep in mind that these DNA sequences are in the context of chromatin. mRNAs produced through this activity are exported to the cytoplasm where they are translated into protein.

2. Transcription factors carry out their role through the following two functions:
   a. The binding of transcription factors to DNA promotes the recruitment and stabilization of the basal transcription machinery (review Dr. Pluciennik's lecture on Molecular Biology Essentials in week 2 of the Intro to Medicine block), beginning with TFIID binding to the TATA sequence (TATA binding protein – TBP- is a component of TFIID) and ending with the recruitment of RNA Polymerase II and transcription initiation.
   b. The binding of transcription factors recruits both coactivators with histone modifying activities (acetylation, methylation, phosphorylation) and chromatin modifying complexes, which together alter the repressive chromatin structure to allow transcription to both initiate and proceed.
      i. Co-activators with histone acetyl transferase (HAT) activity include CREB-binding protein (CBP), and a closely related protein p300 (CBP/p300).
      ii. Chromatin modifying complexes, such as SWI/SNF, contain protein components (e.g., BRG1, as shown in figure below) that bind specifically to modified histones and remodel chromatin in an ATP-dependent manner.
   c. Below is an example of how the glucocorticoid receptor (GR) regulates transcription by binding to a GR response element (GRE).
   d. Nuclear hormone receptors also often function as transcriptional repressors, either directly by binding to DNA or indirectly by forming inhibitory protein-protein interactions with other transcription factors (called transrepression or tethering). Physiological effects of NHR signaling encompasses both transcriptional activating and transcriptional repressive effects (this is particularly true for the glucocorticoid receptor).

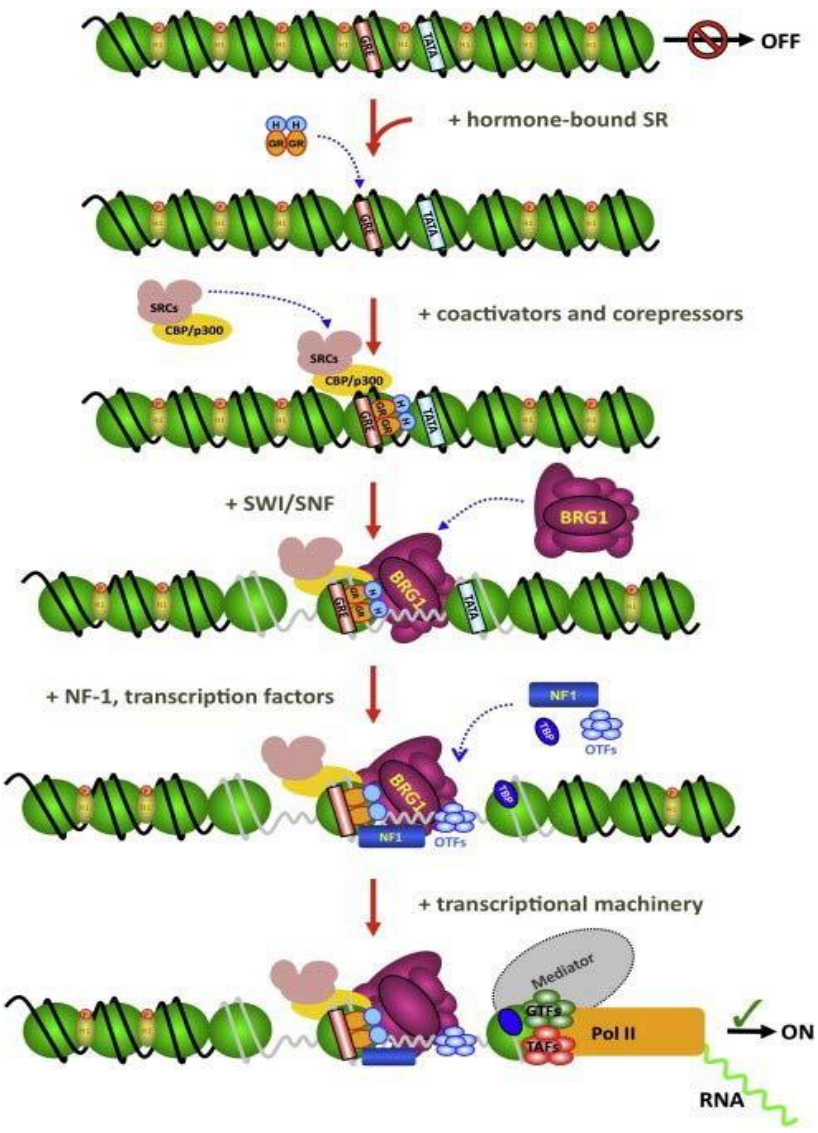


Figure legend. *In the absence of hormone-bound GR, the transcription unit is "off." 2. Cortisol-bound GR binds as a dimer to GRE. 3. The activated GR (hormone-bound) recruits transcriptional co-activators (SRC – steroid receptor coactivator; also CBP/p300). 4. The histone acetylation activity of these co-activators leads to the recruitment of the chromatin remodeling complex SWI/SNF, which contains ATPase activity in the subunit BRG1. 5. The relaxation of chromatin allows the binding of other transcription factors (OTFs), including transcription factor NF1. In addition, with the relaxation of chromatin and exposure of the TATA sequence, TBP (TATA-binding protein) can bind to the TATA promoter sequence. 6. The binding of TBP leads to the further recruitment of the basal transcription machinery, along with Mediator, a large multi-protein complex that is required for the recruitment of RNA Polymerase II. At this point, transcription can begin.*

E. Tissue- and cell-specific effects of NHRs on therapeutic targets

1. Tissue specificity of NHR actions is driven by NHR expression, co- regulator expression, and availability of hormone ligand, which can be controlled by enzymes that synthesize or degrade hormones.

2. Example – MR activity in the nephron.

   You will recall from the lectures in the GI block that one of the regulators of sodium and potassium reabsorption through the nephron is the mineralocorticoid aldosterone. Binding of aldosterone to the mineralocorticoid receptor (MR) activates the transcription of genes involved in kidney reabsorption, including the three subunits of the epithelial sodium channel (ENaC), along with its regulatory kinase, and the Na^+/K^+ pump, along with channel-inducing factor. Conversion of cortisol to cortisone in aldosterone target tissues is critical to ensure MR selectivity for aldosterone.

3. Example – GR activity in the regulation of inflammation.

   You will recall from the Pulmonary block that airway inflammation contributes to the pathology in asthma and COPD. Inhaled glucocorticoids bind to the glucocorticoid receptor (GR) to suppress this inflammation through the transcriptional activation of anti-inflammatory genes and the transcriptional repression of inflammatory genes.

4. Example – AR activity in hair follicles

   a. AR activity is regulated by binding to its androgenic hormones testosterone and the more potent (by approximately 5 times) 5α- dihydrotestosterone (DHT); DHT is produced by the activity of 5α- reductase on testosterone. Hair follicles express high levels of 5α- reductase, resulting in substantial conversion of T to the more potent DHT. High transcriptional activity of AR in hair follicles leads to miniaturization of the follicles, and, over time, androgenic alopecia (also known as male pattern baldness) ensues. Inhibition of 5α- reductase with the drug finasteride (Propecia) reduces DHT levels, which suppresses the miniaturization and loss of hair follicles, thereby stimulating hair growth.



   Figure legend. Male-pattern androgenic alopecia with loss of hair from frontal, temporal and central scalp areas. Contributed by Ahmad Al Aboud, MD. Creative Commons Attribution 4.0 International License.

   b. Androgen insensitivity syndromes (AIS)

      i. Complete androgen insensitivity syndrome (CAIS). Caused by loss-of-function mutations in the androgen receptor (AR) in individuals with a Y chromosome.

a. incidence ~ 1/20,000.

b. underdevelopment of the urogenital structures characteristic of Wolffian development due to inability to respond to adrenal androgens.

c. external genitalia characteristic of Mullerian development, but lacking a uterus.

d. internal testes; infertile.

e. during puberty, internal testes produce testosterone, which is converted to estradiol by the enzyme aromatase. This production of estrogen leads to breast development.

f. diagnosis typically during the time of puberty due to lack of menstruation.

ii. Partial androgen insensitivity syndrome (PAIS). Caused by mutations in the androgen receptor. Partial activation of the androgen receptor results in a spectrum of disease presentation, many of which lead to congenital anomalies of the genitourinary (GU) tract.

Typical presentation includes hypospadias, small penis, and bifid scrotum at birth.

5. Example - Estrogen receptor (ER)

a. Plays a fundamental role in the development of secondary sexual characteristics in XX individuals.

b. ER activities play a central role in progress of menstrual cycle and timing of ovulation. Oral contraceptives consist of a mixture of estrogens and progestin, reversibly inhibiting ovulation by feedback inhibition of gonadotropin release.

c. Two estrogen receptors in humans (α and β); can homo- or heterodimerize (αα, ββ, or αβ).

d. ER is the primary driver of growth and proliferation in hormone- responsive breast cancer.

i. ER antagonists (e.g., fulvestrant) bind competitively to ER (i.e., to the same site as estrogen) and prevent the recruitment of coactivators, thereby preventing ER transcriptional activity.

ii. Selective ER modulators (SERMs) (e.g., tamoxifen, raloxifene) exhibit tissue-specific effects, functioning as an ER antagonist in breast and ovary while functioning as an ER agonist in bone. Useful for treatment of osteoporosis in women at risk for breast or endocrine cancers. Relative levels of coactivators (e.g., SRC) and corepressors (e.g., NCoR) have an

impact on the tissue selectivity of SERMs (see figure below).

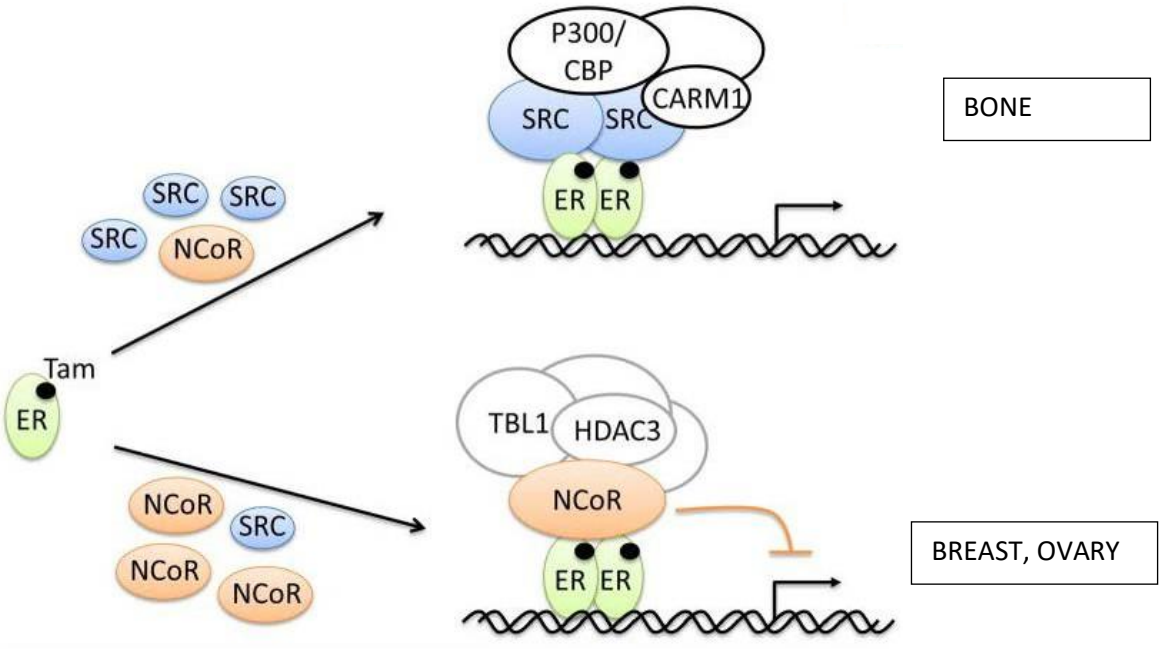


Figure legend. The relative levels of the transcriptional co-activator SRC and the transcriptional co-repressor NCoR determine whether binding of the SERM Tamoxifen (Tam) to the estrogen receptor (ER) leads to transcriptional activation (top; bone) or repression (bottom; breast, ovary). P300/CBP and CARM1 are additional co-activator proteins. TBL1 and HDAC3 are additional co-repressor proteins.

F. Non-steroidal NHR signaling

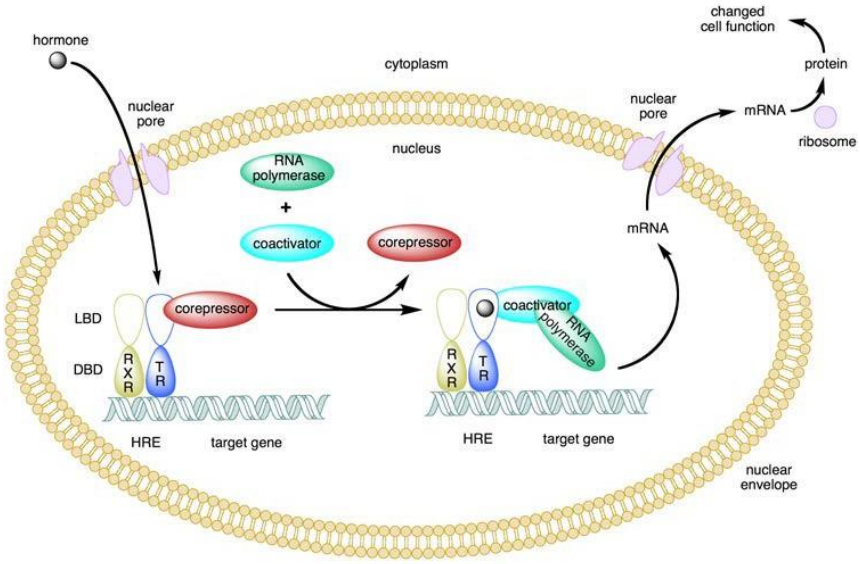
1. Non-steroidal NHR signaling occurs upon the high affinity binding of each non-steroidal hormone or ligand to the ligand binding domain of its specific receptor. The type II NHRs responsible for non-steroidal NHR signaling respond to ligand binding by regulating transcription of specific hormone-responsive genes. The specificity of transcriptional regulation is determined by DNA binding to specific response elements within promoter/enhancer regions of target genes.

2. Thyroid hormone
    a. Thyroid hormone signaling plays an important role in metabolism, including both catabolic and anabolic functions. These include body temperature regulation, metabolic rate, fatty acid synthesis, gluconeogenesis, glycogenolysis and lipolysis. You will learn more about this from the PATH thread. For example, low thyroid hormone (T3) may lead to hypercholesterolemia and the subsequent effect of high cholesterol on cardiovascular function.
    b. Produced by iodinating tyrosine residues in thyroglobulin within specific follicle cells of the thyroid gland.
    c. Most of the thyroid hormone is secreted in the form of T4 (thyroxine).
    d. Some of the thyroid hormone is secreted in the form of T3.
        i. T3 is the major biologically active thyroid hormone.
        ii. Most T3 is generated in peripheral tissues through the action of tissue-specific deiodinases on T4.
    e. Thyroid hormones enter target cells using specific transporter proteins.
    f. Further deiodination inactivates thyronines (e.g., T2, T1, T0).

3. Retinoic acid
    a. Produced from Vitamin A (retinal) by oxidation. Plays critical roles in embryonic development.
    b. Transported to target cells as the alcohol (retinol), bound to RBP4. RBP4 binds to a specific membrane-bound transporter, which transports retinol to the cell interior, where it is converted to retinaldehyde and then to retinoic acid.
    c. You will learn more about retinoic acid signaling in the Dermatology portion of the MSK/Dermatology/Rheumatology block.

G. Non-steroid hormone (Type II) receptors

1. Like the Type I steroidal hormone receptors, Type II nuclear receptors regulate transcription by first binding to specific DNA response elements within gene promoters and/or enhancers according to receptor type. mRNAs produced through this activity are exported to the cytoplasm where they are translated into protein.



Boghog2 - Made by Boghog2, Public Domain, https://commons.wikimedia.org/w/index.php?curid=2375278

2. Type II nuclear hormone receptors form **heterodimers** with retinoid-X receptor (RXR).
   a. Type II nuclear hormone receptors include:

      Retinoic acid receptor

      Vitamin D receptor

      Thyroid hormone receptor

      PPAR α, β/δ, γ*

      (* PPARs have properties of both Type I & Type II receptors; see section H).

   b. Bind primarily to specific DNA sequences within promoters and enhancers of target genes.

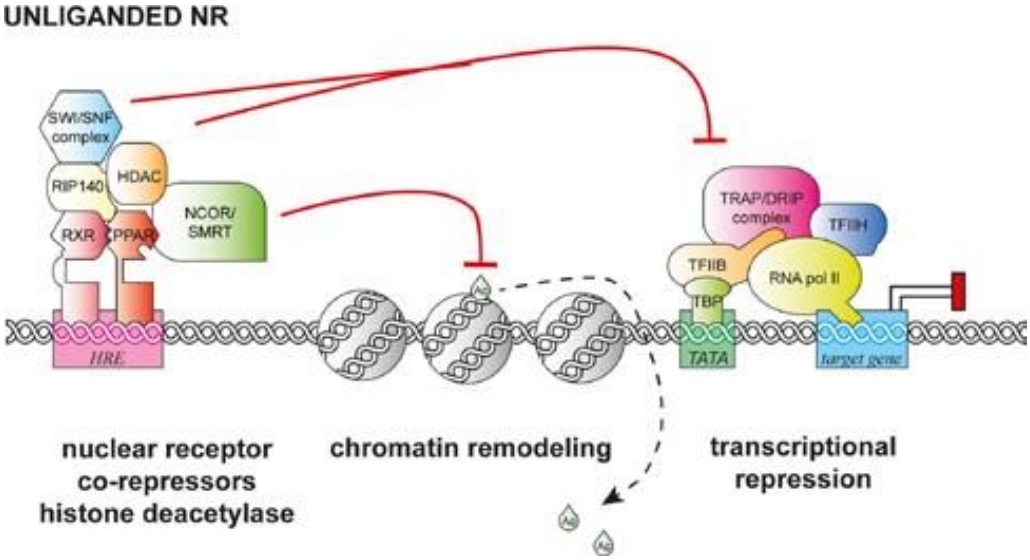
   c. Bind via zinc finger motif within DNA binding domain.

   d. Nuclear and DNA-bound in the absence of hormone.

   e. Activation of Type II receptors

i. Bound to DNA in absence of hormone, but associated with co-repressors. Co-repressors regulate transcription, when bound to DNA-bound transcription factors, by preventing the recruitment of the basal transcription machinery and RNA Polymerase II.

ii. Upon hormone binding, conformational change causes release of co-repressors, and recruitment of co-activators. Co-activators regulate transcription, when bound to DNA-bound transcription factors, by enhancing the recruitment of the basal transcription machinery and RNA Polymerase II and by altering (either directly or indirectly) the compaction



state of chromatin.

Figure legend. Either RXR or PPAR ligand can activate transcription. With VDR, TR, or RAR partners, the VDR, TR, or RAR ligands activate. RXR ligand is 9- cis retinal; RAR ligand is 9-cis retinal and all-trans retinal.

H. PPARs have properties of both Type I and Type II receptors

1. Peroxisome proliferator-activated receptors

   a. Three types: PPAR-α (encoded by NR1C1), PPAR-β/δ (NR1C2), and PPAR-γ (NR1C3).

   b. PPAR signaling is a mixture of canonical Type I and Type II NHR signaling. In the unliganded state, PPARs are mostly cytosolic (similar to type I, steroid-responsive receptors). Upon ligand binding, they heterodimerize with the retinoid X receptor (RXR), translocate to the nucleus and bind to peroxisome proliferator response elements (PPREs). PPAR/RXR heterodimers can either recruit coactivator protein complexes that stimulate transcription, or recruit co-repressor complexes that inhibit transcription (this is similar to canonical

type II receptors). The latter activity is particularly important for PPAR-mediated repression of inflammatory genes.

c. PPARs display overlapping, promiscuous, relatively low affinity ligand binding properties. Physiological effects are mediated by tissue-specific expression profiles of PPARs.

2. PPARα

a. PPARα is expressed primarily in the liver, where it stimulates fatty acid beta oxidation, ketone body synthesis and gluconeogenesis.

b. PPARα is the natural receptor for ω-3 fatty acids and mediates many of their anti-inflammatory effects.

c. PPARα is the target of fibrate drugs, which lower triglycerides and increase HDL (review PHARM lectures from block 3A).

3. PPARγ

a. PPARγ is expressed primarily in adipocytes and skeletal muscle, where it stimulates adipogenesis and increases insulin sensitivity through production of adiponectin.

b. PPARγ is the target of the thiazolidinedione (TZDs, glitazones) class of antidiabetic drugs, which improve insulin sensitivity, lipid profiles and reduce the risk of atherosclerosis.

c. TZDs stimulate renal sodium absorption, which promotes fluid retention and increases risks of heart failure.

Answers to Study Questions

1. Create a table that describes the differences in the mechanism of action of estrogen receptor and thyroid hormone receptor.

Answer:

Estrogen Receptor	*Thyroid Hormone Receptor*
Primarily cytoplasmic in the absence of hormone	*Nuclear in the absence of hormone*
Associated with molecular chaperones (heat shock proteins) in the absence of hormone	*Not associated with molecular chaperones*
Bind DNA in response to hormone	*Bind DNA in the presence and absence of hormone*
Bind DNA as homodimers (to inverted repeat sequence element)	*Bind DNA as heterodimer with RXR (to direct repeat sequence element)*
Recruit coactivators (or sometimes corepressors) upon DNA binding	*Swap corepressors for coactivators in response to hormone binding*

2. The estrogen receptor is a member of the nuclear receptor superfamily and is important for mediating the effects of estrogen in XX and XY individuals. The estrogen receptor, like many receptors in this family, resides in the cytoplasm in the absence of the hormone estradiol and translocates to the nucleus upon binding hormone.

   Once in the nucleus, how does the estrogen receptor lead to transcriptional activation of its target genes? Please start with its binding to DNA and finish with successful transcription initiation by RNA Polymerase II.

   Answer: The estrogen receptor (ER) binds to specific DNA sequences (promoter elements) in actively transcribed genes (and perhaps even in closed chromatin). The ER recruits co-activators with histone acetyltransferase (HAT) activity, as well as coactivators with other histone modifying activities (e.g., methyltransferase, kinase). Modifications are "read" as a "histone code" by chromatin remodeling complexes. The protein complex built by the transcriptional activator serves to recruit and stabilize the binding of the basal transcription machinery, including TFIID, TFIIB, TFIIH, Mediator. TFIIF escorts RNA Polymerase II (Pol II) with Mediator to the complex. TFIIH helicase and kinase activity enables Poll II to initiate transcription.

3. A 9-year-old patient was referred for evaluation of palpable gonads in the inguinal region. At birth, bilateral inguinal gonads with blind-ended vagina were detected. The child was followed over a period of 7 months, after which assignment of female sex was made based on the appearance of external genitalia. Surgical removal of inguinal gonads had been planned, but was not carried out. During the current visit, pelvic sonogram did not reveal any Mullerian structures. Hormonal evaluation revealed FSH (5.4 mIU/mL; normal: 1.5–12.4), LH (21.2 mIU/mL; Normal: 1.7–8.6) and total testosterone (15 ng/mL; normal: 2.86–8.1). Karyotype was 46 XY. Testosterone (T) and dihydrotestosterone (DHT) synthesis defects were excluded by the normal rise of T and DHT after HCG stimulation (basal T: 15 ng/ml; basal DHT: 9.3 ng/ml; T/DHT ratio after HCG stimulation: 7.9; and normal post-HCG stimulation, T/DHT ratio is <17). Surgical removal of inguinal gonads was performed a few months later; histologic analysis revealed normal bilateral testes. The patient subsequently underwent vaginoplasty and received therapy with estrogens. Genomic DNA was extracted and exonic PCR amplification of the androgen receptor (AR) gene performed. Sequence analysis revealed a C to T transition in exon 6 of the AR gene, converting a CAG codon to a TAG codon at the beginning of the ligand-binding domain.

   What is your explanation for the complete androgen insensitivity syndrome presenting in this patient?

   Answer: The C -> T mutation in Exon 6 of the AR gene changes the codon for the amino acid glutamine to a stop codon, prematurely truncating the AR protein and deleting most of the ligand-binding domain. Thus, during development, the truncated AR protein in the Wolffian duct system in this XY individual cannot respond to testosterone produced by the embryonic testes. In the absence of AR response, estrogen receptors in the developing Müllerian duct respond to adrenal estrogens and control development of external genitalia.